

IBM SkillsBuild AI Experiential Learning Lab



Solution Design

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How to Use This Guide

Purpose: This guide offers a template for teams to capture essential information. It aims to help you move from week to week, completing essential exercises to deliver an innovative, user-centric solution at the end of week 10.

Important Notes:

!! Each slide has an indication of by when we recommend that you complete these assignments – however, you are free to advance or delay work according to your schedule.

!! Teams are expected to organized themselves, share tasks, find time to meet and complete the assignments and exercises.

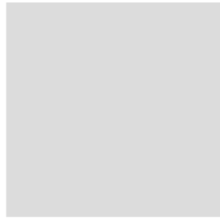
!! Your hard deadline is the end of week 10 –by then, your team must submit a functioning MVP, project documentation and solution demonstration.

!! Submit your weekly deliverables in the forms: <https://wkf.ms/4sDluV8>

Tips:

- ✓ Keep language user-focused, avoid technical jargon, and write in short, clear sentences.
- ✓ Use images and illustrations –one image says more than 1000 words. Copy here images of your boards, designs, etc. or use illustrative images.
- ✓ Customize according to your needs –feel free to change or adapt the templates.

Team Name



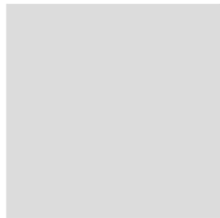
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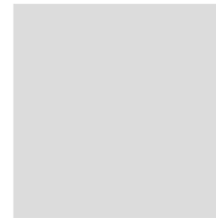
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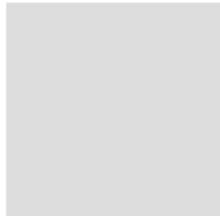
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Opportunity Framing

From industry problem to user-centered opportunity

Before completing the Opportunity Framing section, teams are strongly encouraged to first conduct two foundational exercises:

- Empathy Map –This helps you deeply understand the user’s perspective, emotions, and motivations. It ensures that the opportunity framing is grounded in real user needs rather than assumptions.
- As-Is Scenario Map –This captures the current user journey, including steps, pain points, and context. It provides clarity on where friction occurs and why existing solutions fail.

Completing these exercises first will make the opportunity framing process more accurate and user-centered. They provide essential insights that inform sections such as Core User Tension, Current Experience, and Why Current Solutions Fail.

- Optional: Stakeholder Map –If you are finding it difficult to identify users and stakeholders, this exercise helps capture internal and external actors that might influence your solution.
- Optional: Research Plan -A research plan is essential when starting a project. It fits in especially well right after you identify your team’s Assumptions and Questions.

Industry Problem Context

Purpose: Define the industry-level challenge and why it matters

When: By week 1

Tip:

- Refer to the Industry Problem Statements document.
- As a team, discuss and select a problem statement to tackle.
- In this slide, summarise what you know/understand about the problem. As you learn more and get deeper insights into the industry, update this slide.

Industry challenge statement

Write the official problem statement or summarize it in 1–2 sentences. Be specific and avoid jargon.

Modern streaming viewers are struggling to keep track of increasingly complex, multi-season storylines and massive character networks, leading to narrative confusion and mental fatigue. This "complexity gap" causes many users to stop watching a series entirely because they have forgotten key plot points or feel overwhelmed by the effort required to resume.

Why this problem matters

Explain the impact and urgency from an industry perspective

Address:

- Economic or operational consequences
- Strategic relevance for the industry
- Who is under pressure to solve it (companies, regulators, etc.)

Economic & Operational Consequences:

When viewers feel confused or "lost," they are significantly more likely to abandon a series, leading to lower completion rates and reduced long-term subscription value.

Strategic Relevance for the Industry:

As streaming platforms invest billions into "cinematic universes" and dense lore, the inability of the average viewer to retain that information limits the platform's ability to build lasting, high-value franchises.

Who is Under Pressure to Solve It:

Major streaming giants (such as Netflix, Disney+, and HBO Max) are under pressure to maintain high "user retention" and "session continuation" metrics to justify content spending and satisfy shareholders in a saturated market

Who Is Experiencing the Problem?

Purpose: Identify the people most affected.

When: By week 1

Key tools:

- Empathy Map
- Stakeholder Map

Tip: To gather this information, you will need to carry out industry and user research.

- **Define Clear Objectives:** Start by identifying what you need to learn about = users—e.g., pain points, workflows, or feature expectations.
- **Choose the Right Methods:** Define the best way to gather the required information—desk research, user interview, survey, etc.
- **Document and Synthesize Findings:** Organize the information using key tools and list the key insights about the problem, user needs and pain-points, existing solutions, and opportunities for AI solutions.

Primary user

Answer:

Who directly experiences the friction/pain points? Describe them as a person, not just a job title.

The "Fragmented" Binge-Watcher
This is a person like "Alex," a 28-year-old professional who loves high-quality, complex dramas but has limited free time. Alex often watches an episode late at night after work and might go 10 days before finding time for the next one. Because of these gaps, Alex feels a sense of "narrative dread"—the feeling that they've forgotten too much to enjoy the show, but don't want to re-watch old episodes or risk spoilers by searching online

User environment

Answer:

- Where does the user experience this problem (physical or digital setting)?
- What tools, systems, or processes are currently involved in the user experience?

Physical/Digital Setting:

The problem occurs in the living room or bedroom, typically during "lean-back" entertainment time. Digitally, the user is immersed in a streaming app interface on a Smart TV, tablet, or laptop.

Current Tools & Systems:

Main Platforms: Netflix, Disney+, Prime Video, etc.

Existing Friction Points:

Static "Skip Recap" buttons, low-detail episode descriptions, and external search engines or "Fandom" wikis that often lead to accidental spoilers.

Secondary stakeholders

Answer:

Who else is impacted or influences the outcome? Include decision-makers, support roles, or customers.

Streaming Platform Executives:

Decision-makers who care about "churn rates" and "time spent on platform".
Content Creators & Showrunners: Writers and directors who want their complex stories to be fully understood and appreciated without the audience losing interest.

Advertisers/Sponsors:

Interested in high engagement levels and "completion rates" of the series they are associated with.

Household Members:

Friends or family who might be "further ahead" in a series and have to constantly explain plot points to the primary user, creating a disrupted group viewing experience.

Core User Tension

Purpose: Capture the heart of the opportunity

When: By week 1

User goal

1 sentence: What is the user trying to achieve? Keep it short and clear.

The user is trying to enjoy a complex, episodic story with full comprehension and emotional engagement, despite having long gaps between viewing sessions.

Current User Experience

As-Is Snapshot

Map the existing journey and key steps users take now

Key pains / frustrations

Max 3: List the top fiction/pain points that block the user's goal.

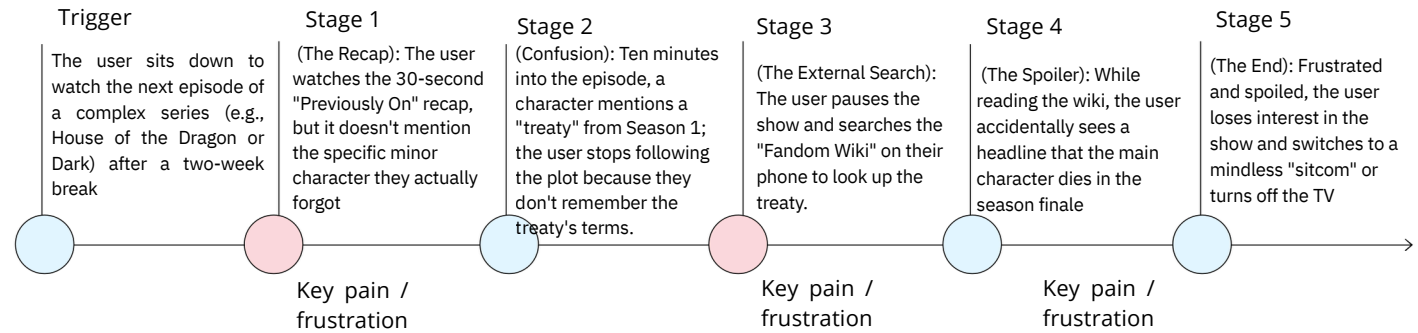
- 1 Cognitive Overload: The mental effort required to keep track of nonlinear timelines and massive character networks leads to "viewing fatigue."
- 2 Narrative Disconnect (Context Loss): Existing recaps are static and "one-size-fits-all," failing to address the user's specific confusion or memory gaps
- 3 The "Spoiler Minefield": Using external tools (Google, Wikis, YouTube) to find answers creates a high risk of accidental plot spoilers that ruin the emotional experience.

Key tools:

- Empathy Map
- As-Is-Scenario

Tip: Focus on understanding what users are experiencing.

- Interview or survey users
- Consult with industry experts
- Explore industry reports and case studies



Cognitive Overload:

Narrative Disconnect
(Context Loss):

The "Spoiler Minefield"

Where friction (pain points)
occurs

Highlight where pain-points, such as delays, errors, rework, or cognitive overload happen.

Friction occurs most heavily during Stage 3 (The External Search) and Stage 4 (The Spoiler). This is where the user's immersion is broken, their cognitive load shifts from "enjoying" to "researching," and they risk "rework" (having to re-watch old episodes) just to understand the current one

Source: If applicable, describe source origin

Why Current Solutions Fail

Purpose: Understand gaps in existing tools/processes

When: By Week 1/2

Key insight – why the problem exists

What makes it hard to solve the problem today?

Complexity, ambiguity, unstructured data, coordination, etc.

Explain why “more automation” or “better UI” alone won’t solve the problem.

Key tools:

- Empathy Map
- As-Is-Scenario

Tip: Focus on understanding what solutions users are already using and why they don’t solve the problem – this will help you understand what opportunities exist for you to innovate and offer better solutions.

- Research existing solutions
- Interview or survey users
- Consult with industry experts

The fundamental challenge isn’t just a lack of information, but the lack of contextual awareness. Solving this is difficult because:

- Narrative Complexity & Ambiguity: Modern stories use non-linear timelines and massive, overlapping character networks that are difficult for static systems to track.
- Unstructured Data: Story details are buried in hours of audiovisual content, making it a “big data” problem to find one specific relationship or plot point.
- Why UI/Automation Fails: Simply adding “better UI” (like a better menu) or “more automation” (like auto-generated subtitles) doesn’t solve the problem because those tools don’t understand the story’s meaning or where the user is in their viewing journey.

Existing tools / processes used today

List what solutions users rely on now today

Users currently rely on a fragmented mix of manual and static tools:

- Built-in Platform Recaps: The “Previously On” video segments at the start of episodes.
- External Wikis & Databases: Websites like Fandom, Wikipedia, or IMDb.
- Social Media & Forums: Reddit threads or YouTube “explained” videos.
- Manual Scrubbing: Rewinding or fast-forwarding to previous episodes to find a specific scene.

What they do well

Identify strengths of existing solutions to avoid reinventing the wheel

To avoid reinventing the wheel, we must acknowledge the strengths of these current methods:

- Recaps: Provide a high-level emotional “vibe” and remind users of the most major plot twists.
- Wikis: Offer incredibly deep, community-verified lore and exhaustive character biographies.
- Platforms: Are excellent at high-quality content delivery and basic metadata (titles, cast lists).

Where they break down

Explain why this solution fails to solve the user pain points (e.g., too manual, fragmented, rigid)

These solutions fail because they are manual, fragmented, and rigid:

- Static & Non-Adaptive: A video recap is the same for everyone; it doesn’t care if you specifically forgot who the “King’s Hand” is while remembering everything else.
- The Spoiler Risk: External wikis are “future-aware.” A user looking for a Season 1 detail will often accidentally see a “Date of Death” from Season 5, ruining the experience.
- High Friction: Forcing a user to pause their TV, pick up a phone, and search Google completely breaks narrative immersion and “flow”

Early AI Opportunity Hypothesis

Purpose: Understand opportunities to offer

better solutions (No solution design yet)

When: By Week 1/2

Why Agentic AI?

Answer:

- What would the system understand better with Agentic AI?
- What decisions could be improved?
- What coordination could be simplified?
- Deep Contextual Understanding: Unlike standard LLMs, an agentic system can maintain a persistent "Narrative State," understanding not just words, but the evolving relationships, motives, and temporal shifts within a fictional universe.
- Improved Decision-Making: The AI can make real-time decisions about "Information Disclosure"—determining exactly what a user needs to know to resolve confusion while proactively filtering out any data that constitutes a future spoiler.
- Simplified Coordination: The system simplifies the coordination between multimodal inputs (video, audio, subtitles) and the user's specific "knowledge gap," acting as a bridge that aligns the current scene with the viewer's personal memory of past events

Key tools:

- Empathy Map
- As-Is-Scenario
- Big Idea Vignettes

Tip: Focus on understanding how AI can offer value beyond basic prompting and automation –this will help you understand what opportunities exist for you to innovate and offer better solutions.

- Understand Agentic AI capabilities
- Consult with industry experts
- Research innovative Agentic AI solutions in other areas/industries that might give you inspiration

High-level hypothesis

1–2 sentences - Use this format:

“An intelligent system could help users by [what it would do], because today [why current approach fails].”

“An intelligent system could help users by [what it would do], because today [why current approach fails].”



No features. No tools. No architecture.

"An intelligent system could help users by restoring narrative continuity and contextual comprehension in real-time, because today viewers face high cognitive loads and spoiler risks when trying to manually bridge gaps in complex, long-form storytelling



Checkpoint: Week 2

At this stage, you should have a clear picture of the problem, the people affected, and why current solutions fail —without jumping into solution design.

- ☐ The problem is specific and human-centered (experienced by real people)
- ☐ The user and user experience is clear
- ☐ The pain-points are solvable and non-obvious
- ☐ There is a clear and valuable opportunity for Agentic AI to offer a better solution and experience

Key tools:

- Empathy Map
- As-Is-Scenario

Tip: There are things we know —there are things that we think we know (assumptions) —and there are things we do not know. Now it is the time to look again at your empathy map and as-is-scenario and do a critical analysis. Is there something that is critical for the success of your project that you might not know for sure yet?

User value is at the heart of innovation. The market is full of solutions that solve no-one's problem because they failed to understand their user needs, pain-points and desires —this is the time for you to be sure that you are solving a real, human problem, that will make users want to use your solution.

Open Questions & Assumptions

What we don't know yet

List gaps in knowledge that need validation.

- API Accessibility: We do not yet know the extent to which major streaming platforms (e.g., Netflix, HBO Max) allow external third-party applications to access real-time playback data or "current timestamp" metadata.
- Narrative Modeling Depth: It is unclear how deep the "Narrative Graph" needs to be —specifically, how to mathematically define the boundary between a "helpful reminder" and a "minor spoiler" for different genres.
- Latency Constraints: We need to determine the maximum acceptable delay between a user asking "Who is this?" and the AI providing a response without breaking the viewer's immersion.

Assumptions we are making

Document any assumptions about users, processes, or technology.

- User Behavior: We assume that users are willing to interact with a "second screen" (phone/tablet) or a TV overlay while watching, rather than just being passively confused.
- Technology Readiness: We assume that current Multimodal LLMs are sophisticated enough to perform "temporal reasoning" (tracking changes over time) across multiple seasons of content.
- Data Availability: We assume we can source or generate high-quality "ground truth" narrative data (scripts, lore wikis, scene descriptions) to feed the Retrieval-Augmented Generation (RAG) system.

Validation research

List what research is needed to gather information and/or validate assumptions

- [] Technical Feasibility Audit: Conduct a review of Chrome Extension capabilities and Smart TV SDKs (like Tizen or webOS) to see if we can overlay the companion directly onto the video stream.
- [] User Experience (UX) Survey: Interview "Binge-Watchers" to identify the "Drop-off Point"—how many episodes of confusion usually pass before they decide to abandon a series.
- [] Accuracy Benchmarking: Run a pilot test using a complex show (like Dark or Game of Thrones) to see if the AI can correctly identify "spoiler-safe" information for a viewer who is only halfway through Season 2

User-Centric Solution

From opportunity to concept— addressing user pain-points

Before completing the User-Centric Solution section, teams are strongly encouraged to first conduct two foundational exercises:

- Ideation

✓ Tip: Keep language user-focused, avoid technical jargon, and write in short, clear sentences.

Target Core User

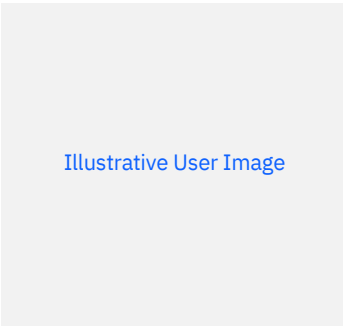
Purpose: Ensure that your solution is user-centered and solves real user problems
When: ByWeek3

Key tools:

- User Persona

Tip: Once you are Summarise the information

Targeted User Persona



Firstname Lastname
Role

Demographic info
Other demographic information, such as age, gender, income, education, location, etc.

Short Bio Explain who is the user, their job description, their preferences, and any other background information can help you create a more authentic character. Name: Alex Demographics: 28 years old, urban professional. Job Description: Project Manager in a fast-paced tech firm. Background & Preferences: Alex has a high-pressure job and uses streaming as a primary way to decompress. He prefers "prestige TV"—complex, high-stakes dramas and sci-fi series with deep lore—but his schedule is unpredictable. He often has to pause a series for weeks due to work deadlines. He is tech-savvy but values "lean-back" entertainment and dislikes having to "work" (searching wikis) to understand a show.	Goals & Motivations What your audience wants to get or to do with your product. List all possible goals and motivations that you've found from your user research. - Narrative Continuity: To resume a complex show after a long break without feeling lost or needing to re-watch previous episodes. - Deep Engagement: To understand the subtle political shifts and character motivations that make high-quality storytelling rewarding. - Spoiler-Free Exploration: To satisfy curiosity about a character's history or a plot detail without accidentally seeing future "death spoilers" on the internet. - Efficiency: To get immediate, context-aware answers while staying "in the moment" of the viewing experience	Frustrations & Pain Points List out all possible problems, challenges, and even fears that your audience may encounter while interacting with your products. This information can help you create a product that really meets the needs of your users. - The "Recap Gap": Standard "Previously On" segments are too brief and often don't cover the specific minor details Alex has forgotten. - Cognitive Overload: Feeling mentally exhausted by trying to remember non-linear timelines or massive casts of characters. - Immersion Breaks: The frustration of having to pause the TV to look something up on a phone, which ruins the cinematic "flow". - Spoiler Anxiety: A genuine fear that searching for a simple "Who is this?" will lead to a search result that ruins the entire season's climax.
Ideal Experience / Must Have Knowing your core user, explain how your product can serve them better to create the optimal user experience based what they would love to or must have. - Contextual Awareness: The solution must know exactly which episode and minute Alex is watching to provide relevant information. - Spoiler-Safe Guardrails: The AI must filter its knowledge base to hide any information that occurs after the current timestamp. - Non-Intrusive Interface: Ideally, the experience is a "second-screen" companion or a subtle overlay that doesn't block the actual video content. - Natural Language Interaction: Alex should be able to ask questions in plain English (or his native language) rather than using complex search queries.	Skills & Capabilities List the industry and AI technical capabilities that your core user has—including potential gaps that your solution will have to consider. - Industry Knowledge: High familiarity with streaming platforms (Netflix, HBO, etc.) and digital interfaces. - Technical Literacy: Proficient with smartphones and basic AI tools like ChatGPT, though he may not understand "Agentic AI" or "RAG" backend logic. - Capability Gaps: - Limited "Deep Lore" Memory: While Alex is smart, he lacks the time to memorize character family trees or fictional histories. - Focus Fragmentation: He often multi-tasks or watches while tired, leading to decreased information retention. - No Technical AI Debugging Skills: He expects the system to "just work" and will likely abandon the tool if it provides incorrect or hallucinated narrative facts.	

Solution Concept (Part 2)

Purpose:Understand what your solution
will do and how it will solve the user
pain-points When:By Week 3

Key tools:

- Big Idea Vignettes
- To-Be-Scenario

Tip: ...

Idea Cluster Play Back

List the top ideas the team wants to pursue

The team will pursue three core "pillars" of the system to address user confusion:

The Spoiler-Safe Vault (Dynamic Narrative Masking):
An intelligent filter that uses temporal reasoning to ensure the AI only "knows" what the user has watched.

Multimodal Context Awareness:

A system that processes visual (faces), audio (dialogue), and subtitle data to know exactly who is on screen at any given moment.

Active Continuity Restoration (Memory Core):

A retrieval-augmented system that bridges the gap between viewing sessions by maintaining a "Narrative State Engine" for the user

Value Proposition & AI Justification

Explain why these ideas will solve user pain-points and deliver user value through agentic AI.

Why would this solution be interesting even if the AI technology improves tomorrow?

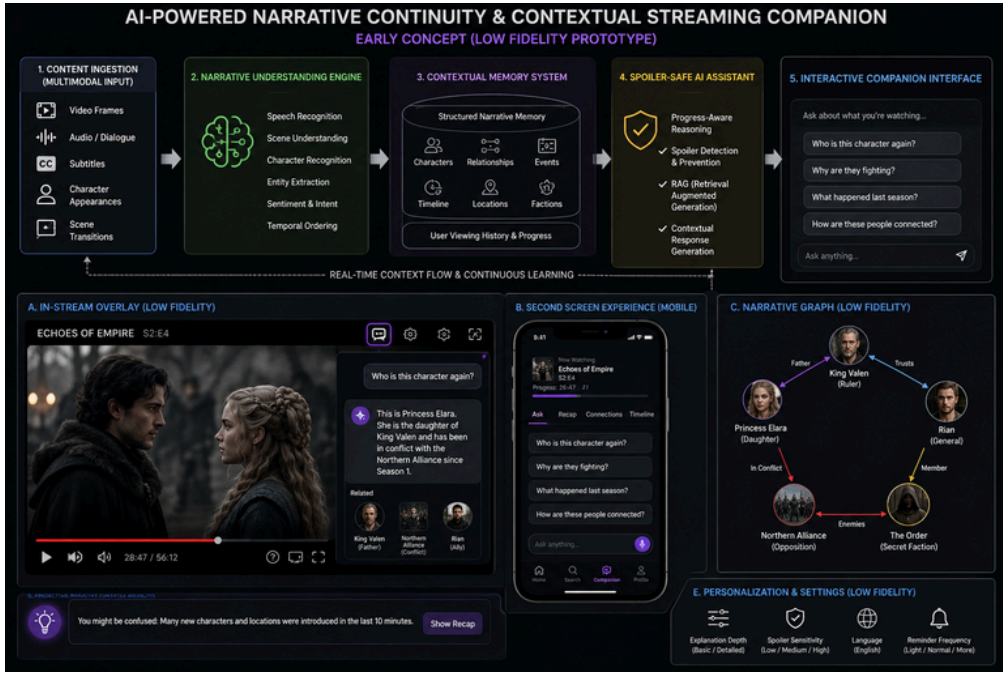
Solving User Pain Points

- Cognitive Load Reduction: By using Agentic AI to manage the "Narrative Graph," the user no longer has to mentally track 50+ characters or complex timelines.
- Eliminating Immersion Breaks: The agent proactively identifies moments of high confusion (e.g., rapid scene switching or new character introductions) and offers lightweight clarifications, preventing the need for external searches.
- Guaranteeing Spoiler-Safety: Unlike generic search engines, this system uses temporal reasoning to strictly separate "current context" from "future plot," removing the anxiety of accidental discovery.

- Future-Proof Interest
- Domain-Specific Reasoning: Even as general AI models improve (becoming faster or cheaper), the specialized narrative logic and the proprietary "Narrative State Engine" developed for this project remain unique assets.
 - Ecosystem Integration: The value lies in the integration layer between AI intelligence and streaming platform data; as AI improves, the companion simply becomes more accurate and faster, but the core need for "narrative assistance" in complex storytelling will only grow as content becomes more dense

Ealy Concept (Low Fidelity Prototype)

Summarize/Illustrate here how your ideas would come together as a powerful and innovation agentic AI solution.



Innovation Intent

Purpose: Explain why this solution is different
When: ByWeek3

User goal

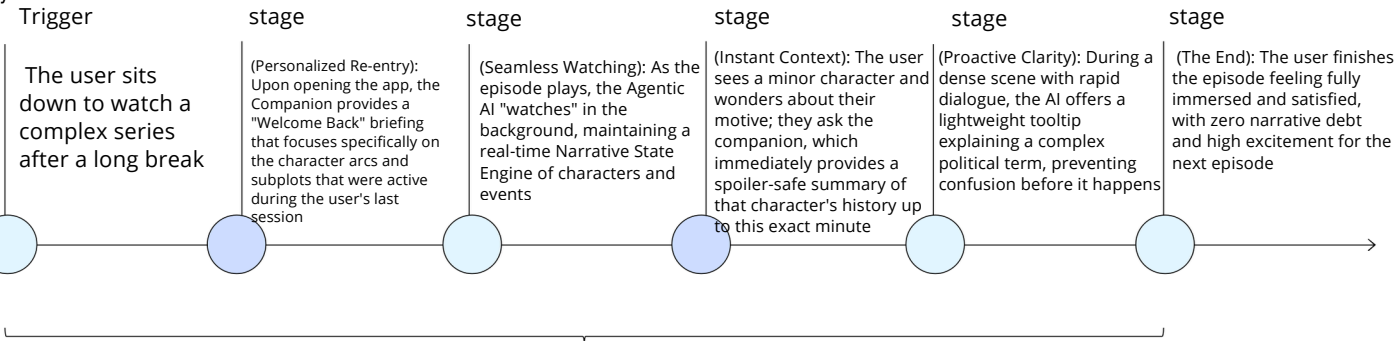
1 sentence: What is the user trying to achieve? Keep it short and clear.

The user is trying to maintain full narrative comprehension and emotional engagement during complex, multi-season storytelling without experiencing cognitive overload or accidental spoilers

Proposed transformation

In one paragraph, describe how the experience changes with your solution

With the AI Narrative Companion, the viewing experience shifts from a passive, often confusing activity into a context-aware, interactive journey. Instead of pausing the show to risk spoilers on a search engine or feeling "lost" after a long break, the user has a seamless intelligence layer that bridges their personal memory gaps. The experience becomes "frictionless" as the system proactively handles the mental heavy lifting of tracking character arcs and complex timelines, allowing the viewer to stay fully immersed in the story's emotional flow



Why this solution

Explain why this Agentic AI solution is better than existing solutions.
What is the core value of AI? How does it go beyond simple prompting or automation?

This Agentic AI solution is superior to existing methods—like static recaps or fan wikis—because it possesses temporal reasoning and contextual memory.

- Beyond Simple Prompting: Traditional AI requires the user to know exactly what to ask and risks hallucinations; this agent "watches" the content alongside the user, understanding the visual and narrative state in real-time to provide grounded, factual answers.
- Beyond Basic Automation: While automation can generate subtitles, this agent makes autonomous decisions about what information is "safe" to reveal, acting as a dynamic filter that prevents future plot points from being disclosed.
- The Core Value: The AI serves as a Narrative State Engine, transforming unstructured audiovisual data into a structured knowledge graph that is personalized to the user's exact progress and confusion level.

Key tools:

- Big Idea Vignettes
- To-Be-Scenario

Tip: ...

What makes this innovative?

New workflow
New decision-making logic New coordination model New use of data/knowledge New level of autonomy

- New Workflow: The system introduces a "Second-Screen" or "Overlay" Interaction Model. Instead of the user pausing the media to perform an external search (breaking immersion), the workflow becomes a seamless, parallel stream where the AI "watches" with the user and provides instant, non-disruptive clarifications.
- New Decision-Making Logic: The agent utilizes Temporal Reasoning to determine "Information Eligibility". It makes autonomous decisions on what to reveal based on the user's specific timestamp, ensuring that every response is strictly "spoiler-safe" by hiding future plot points.
- New Coordination Model: This solution coordinates Multimodal Data Streams (video, audio, and text) with a personal Narrative State Engine. It aligns what is currently happening on screen with what the specific user remembers (or has forgotten), bridging the gap between the digital content and human memory.
- New Use of Data/Knowledge: It transforms Unstructured Audiovisual Data into a structured, queryable Narrative Knowledge Graph. This allows the system to understand complex relationships, political factions, and lore as "data points" rather than just pixels on a screen.
- New Level of Autonomy: The companion operates with Contextual Independence, proactively identifying "confusion hotspots"—such as rapid scene changes or the introduction of many new characters—and offering help before the user even has to ask.



Checkpoint: Week 3

At this stage, you should have a clear picture of what your solution will look like – this is your first solution design.

- ☐ Is the concept clear?
- ☐ Is this solution different/innovative compared to what is already in the market?

Key tools:

- Ideation
- Concept prototyping

Tip: Early testing & validation Start getting use and expert feedback by showing your solution concept.

Open Questions & Assumptions

What we don't know yet

Narrative Granularity:

We need to determine the exact mathematical threshold for "spoiler detection"—specifically, how the AI distinguishes between a "needed reminder" and a "future plot twist" across different genres.

Multimodal Latency:

We do not yet know the hardware requirements for real-time video/audio processing to ensure the AI doesn't lag behind the streaming content.

Streaming Content Rights:

We need to validate if "fair use" or streaming APIs allow the companion to overlay information directly on top of copyrighted video streams

Assumptions we are making

User Interaction:

We assume viewers are willing to trade a small amount of screen real estate (or use a second screen) in exchange for narrative clarity.

Context Persistence:

We assume that LLMs can maintain a consistent "long-term memory" of a fictional universe that spans dozens of hours of media.

AI Factuality:

We assume the "Narrative State Engine" can prevent hallucinations, as providing a single "fake" plot point would destroy user trust

Validation research

Prototype User Testing:

Conduct a "Wizard of Oz" test where a researcher acts as the AI to see if the "spoiler-safe" answers actually reduce cognitive load for viewers watching a complex pilot episode.

Technical Benchmark:

Run a series of "stress tests" on multimodal models to measure the accuracy of character recognition in dark or high-action scenes

Market Comparison:

Analyze existing "X-Ray" features (like Amazon Prime's) to identify the specific "gap" where our Agentic AI provides value that static metadata cannot

Solution Blueprint

From idea to reality –
building strong AI solutions

AI Agent Architecture Purpose:Show the structural designof your agent –what major components it consists of and how they connect. When:By week 4-5

- Key tools:
- IBMOrchestrate

Tip: Architecture determines what your agent can do and how it does it.

- ✓ Use clear diagrams (e.g., flowcharts for workflow).
- ✓ Keep text concise and diagrams labelled.
- ✓ Be ready to explain each component and transition.

Agent Overview

The platform acts as a context-aware narrative intelligence layer that monitors streaming content in real time to provide personalized, spoiler-safe continuity support and relationship mapping for viewers of complex media

Agent roles (single or multiple):

This system utilizes a multi-agent architecture to separate the high-compute tasks of "watching" the content from the logic-heavy tasks of "reasoning" about the story

Agent	Responsibility	Key Decisions
Agent 1: Perception agent	Processes multimodal streams including video frames, audio, and subtitles to recognize active entities on screen.	Identifies context: Decides which characters, factions, or objects are currently "in focus" during the playback.
Agent 2: Narrative Archivist	Manages the Narrative Knowledge Graph and queries the historical memory store for relevant plot data.	Contextual Retrieval: Decides which past events or relationships are most relevant to the current scene's confusion points.
Agent 3: Governance Agent	Acts as the "Spoiler Shield" by comparing retrieved data against the user's current progress metadata.	Information Disclosure: Critically decides which information is "safe" to show and which must be filtered to prevent spoilers.
Agent 4: Interaction Agent	Synthesizes the final response or UI tooltip into a clear, non-intrusive format for the user.	Delivery Logic: Decides whether to provide a short text reminder, a relationship diagram, or a proactive "clarification tip".

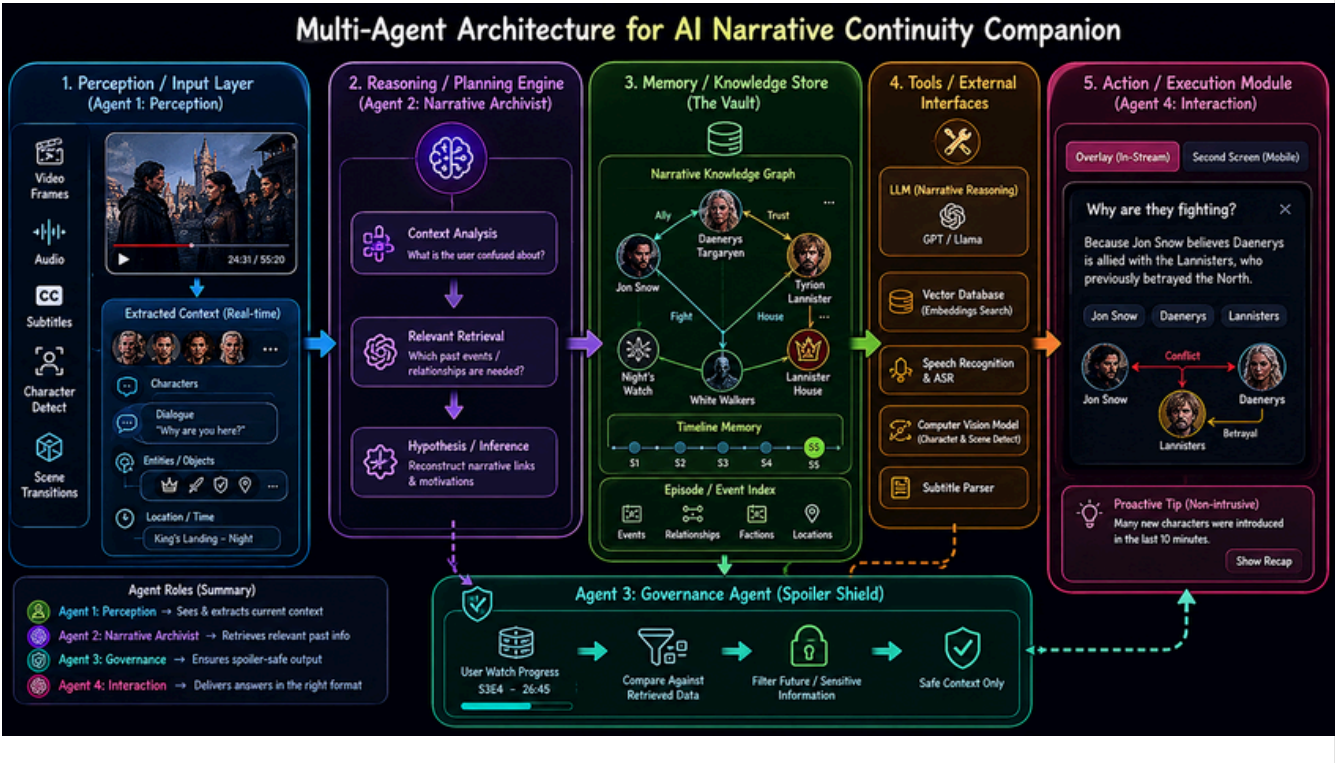
Why this architecture?

Separation of Concerns: By isolating "Perception" from "Governance," we ensure that the system's "vision" doesn't accidentally leak future plot points into the "reasoning" layer.

Scalability: A modular design allows for independent updates; for example, the Perception Agent can be upgraded to a better computer vision model without needing to rewrite the Narrative Knowledge Graph.

Reliability: Decoupling ensures that if the UI/Interaction layer lags, the core Narrative State Engine remains accurate and synchronized with the video timestamp

Architecture Diagram



Tracing Data Flow: From Input to Action

- The flow of data is a linear progression through specialized nodes to ensure accuracy and safety:
- Input Layer (Perception):** Raw audiovisual data (pixels, audio frequencies, subtitles) is ingested. This component is necessary because the AI must "see" what the user is seeing to provide relevant context.
 - Reasoning/Planning Engine:** The Perception Agent identifies entities and passes them to the Narrative Archivist. This engine acts as the "brain," deciding what information from the vast series history is actually relevant to the current frame.
 - Memory/Knowledge Store (The Vault):** The agent queries a Vector Database containing the "ground truth" of the show's lore. This is necessary to prevent hallucinations and provide factual, grounded answers.
 - Governance Layer (The Spoiler Shield):** Before the data reaches the user, the Governance Agent filters it against the user's "Watch-Point Metadata". This is the most critical design decision, as it prevents accidental spoilers that would ruin the user experience.
 - Action/Execution Module:** The Interaction Agent packages the safe information into a user-friendly UI (overlay or second screen). This is necessary to deliver the solution in a "lean-back" manner that doesn't break immersion.

AI Agent Workflow Purpose: Show the dynamic process your agent follows to achieve its goals —step by step.
When:By week 4-5

Key tools:
• IBM Orchestrate

Tip: Ensure that the sequence clear and logical and that iterations and decision points are indicated indicated.

- ✓ Use clear diagrams (e.g., flowcharts for workflow).
- ✓ Keep text concise and diagrams labelled.
- ✓ Be ready to explain each component and transition.

Workflow Description

Ashortexplanationoffhowtasks are handled

The workflow follows a Multimodal-to-Narrative pipeline. When a user triggers a query (via voice or button), the Perception Agent captures the current screen state. This is passed to the Narrative Archivist, which retrieves lore from the Memory Store. The Governance Agent then validates the data against the user's progress before the Interaction Agent delivers the response

Decision Points

Explainwhereand how agents make decisions -what conditions make the agent branch or retry?

The agent branches its logic based on Confidence Thresholds and Watch-Point Progress:

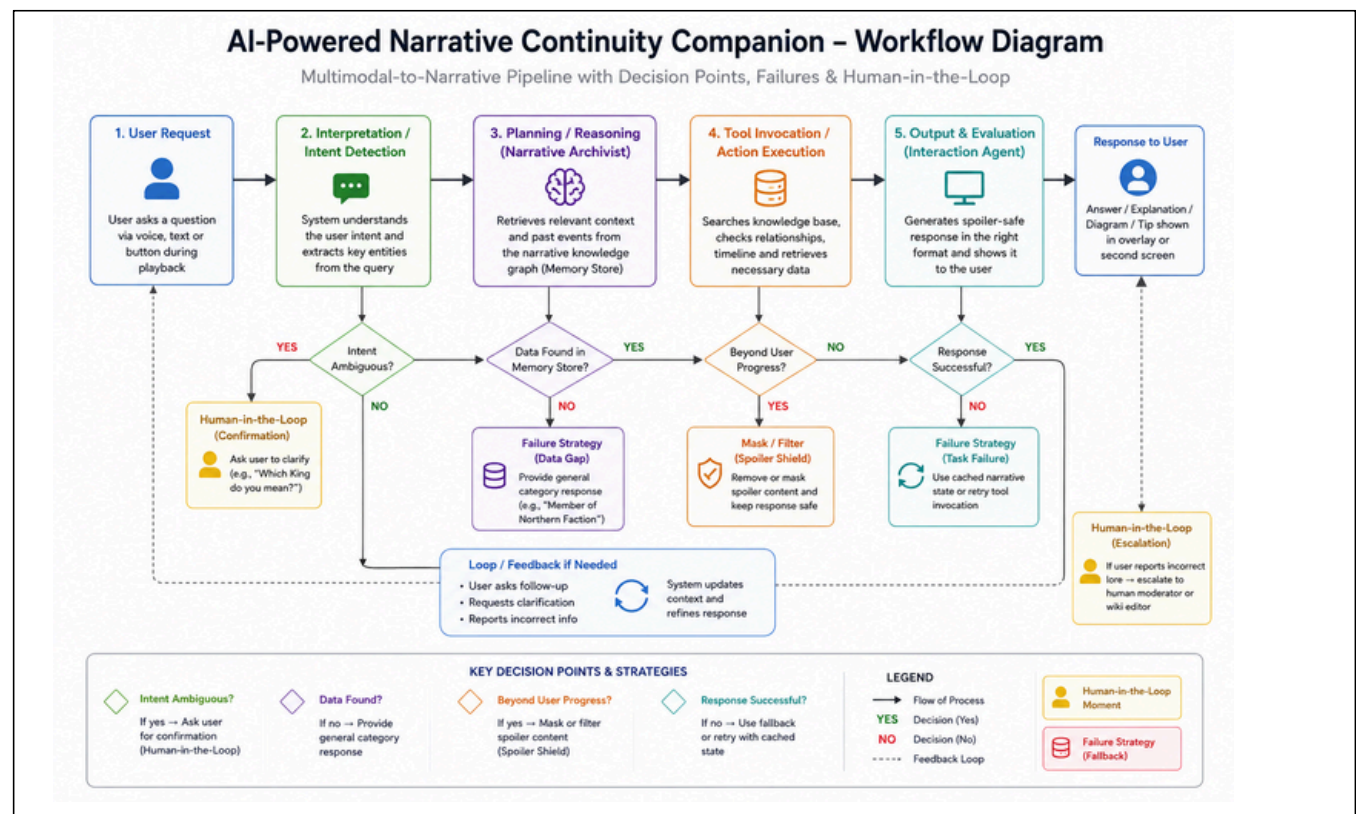
- Narrative Branching: If the retrieved lore is tagged with a "timestamp" greater than the user's current progress, the agent branches to a "Masking" logic to hide the spoiler.
- Retry Logic: If the Perception Agent cannot identify a character with >85% confidence (due to a dark scene or costume change), it retries using the Audio/Subtitle stream for cross-verification.

Human-in-the-loop moments

When does theagent ask forconfirmation?
When does it escalate to a human?

There are specific triggers where the AI seeks human intervention to build trust:

- Confirmation: The agent asks for confirmation if the user's voice command is ambiguous (e.g., "Do you mean the King of the North or the King of the East?").
- Escalation: If a user reports a "Incorrect Lore" error, the system escalates the query to a Human Content Moderator or logs it for a community "Wiki Editor" to review the Narrative Graph accuracy.



Failure strategy nodes

What happens if theagentlacks data? What happens if a task fails?

To prevent the system from "breaking the fourth wall" with errors, we use specific fallback strategies:

- Data Gaps: If the Memory Store lacks data on a specific character, the agent provides a general category response (e.g., "This character is a soldier of the Northern Faction") rather than hallucinating.
- Task Failure: If a real-time API call to the streaming platform fails, the system falls back to its cached Narrative State from the start of the episode to maintain a best-guess continuity.

Tools & Integrations Purpose:
List the external systems your agent uses — APIs, data sources, models, or libraries. When: By week 4-5

Key tools:

- IBMOrchestrate

Tip:You can combine any supported external tool & integration to create and/or enhance your agentic AI solution. Keep in mind: Are the tools necessary and justified? How does integration affect performance?

Tools List & Purpose

Tool / Service	Used For	Integration Type	Executables vs. Autonomy
watsonx.ai APIs	Core Intelligence: Running text inference and prompt tuning for narrative reasoning.	REST API	Automated Action: Invoked by the reasoning engine to process queries.
watsonx.data (Milvus)	Memory Store: Storing and querying vectorized embeddings of show lore for RAG.	Embedded Vector DB	Automated Action: Seamlessly retrieved by the agent to ground responses in trusted data.
watsonx Orchestrate	Agentic Workflow: Coordinating execution across the perception, memory, and governance agents.	Orchestration Engine	Autonomous: Plans and executes sequences to achieve the user's goal.
Streaming Platform API	Synchronization: Syncing the companion with the user's current episode and timestamp.	REST / Webhooks	Automated Action: Sync happens in background; user verifies if it fails.
Custom UI (Extension/App)	Interaction Layer: Displaying spoiler-safe overlays or mobile companion cards.	Web / Mobile SDK	User-Verified: The viewer views and interacts with the AI's provided context.

Notes on Constraints

1. Perception Latency: Running heavy computer vision models on raw 4K streaming video is too slow and expensive. To resolve this, user devices will downsample video frames locally (converting heavy images into small 224x224 feature maps) before shipping them to the cloud.
2. watsonx Quotas: Message routing inside watsonx Orchestrate must be structurally optimized to stay within session caps, caching standard relationship graphs locally to avoid redundant API hits.
3. Data Security & Privacy: Audiovisual frames captured from a user's TV must never be stored permanently or used for external training, complying with standard user privacy frameworks.

Optional Intelligence & Reasoning Design
Purpose: Define the technical depth –the brains of the agent apart from just architecture and workflow. When: By week 5

Knowledge & Retrieval	Reasoning & Decisions	Multi-Agent Logic (if used)	Risks & Constraints
- Knowledge Domain: Structured narrative graphs (alliances/lineages) and unstructured lore (scripts/wikis). - RAG Strategy: - Sources: watsonx.data (Milvus) Vector DB for semantic search; Relational DB for episode facts. - Retrieval Logic: Hybrid search using Temporal Filtering. Metadata tags (revealed_in_episode) exclude all data beyond user’s current timestamp. - Accuracy Control: Metadata-driven pre-filtering prevents future plot points from entering the context window.	- Key Decisions: Intent classification (ID vs. Motivation) and Information Eligibility (Spoiler check). - Reasoning Patterns: Multi-step Chain-of-Thought (CoT): Sense context → Map relationships → Scrub for spoilers → Formulate response. - Outcome Evaluation: 85% confidence threshold for visual identification; structured JSON validation for UI consistency.	- Why Multiple Agents: Separates "Perception" (compute-heavy) from "Reasoning" (logic-heavy) to prevent data leaks and optimize latency. - Interaction Pattern: Sequential pipeline (Perception → Archivist) with Supervisory Governance Agent (The Spoiler Shield) auditing all final output.	- Technical Risks: Inference latency. Mitigation: Local frame downsampling and async pre-fetching of anticipated lore. - Data & Model Risks: Canonical hallucinations. Mitigation: Zero-temperature settings and strict RAG grounding (system stays silent if data is missing). - Autonomous Tradeoffs: Proactive vs. On-Demand help. Mitigation: User-adjustable intervention levels to protect immersion.

Governance & Ethics Purpose: Explain how your AI agent is designed to behave safely, responsibly, ethically and in compliance with privacy and governance expectations. Include mechanisms that prevent harm, protect users, and build trust.

When: by Week 5

Ethical Considerations

1 - Ethical Considerations

a) Fairness & Anti-bias

- Strategies to detect or avoid biased decisions: Ensure the system treats all storylines and characters equally, without prioritizing specific cultures, genders, or genres when summarizing facts.
- Testing plans or mitigation steps for bias: We will test the system across a diverse range of global shows (e.g., international dramas, anime, diverse casts) to ensure accuracy is consistent everywhere.

b) Transparency & Explainability

- How outcomes can be explained to users: Every answer will include a simple source tag (e.g., "Based on Season 1, Episode 3"), so the user knows exactly where the information came from.
- Audit logs or decision traces: The system will keep temporary, private logs of the reasoning steps. If a user reports an error, developers can trace exactly why the AI chose to show that specific information.

c) Accountability & Human Oversight

- Who is responsible for decisions the agent makes? The development team is ultimately responsible for the system's rules and safeguards.
- What happens if something goes wrong? If the AI hallucinates or reveals a spoiler, the user can hit a "Flag Error" button. Repeated flags will automatically pause the AI for that specific show until human moderators review and fix the lore database.

d) Respect for Rights

- Avoiding discriminatory or harmful practices: The AI is restricted to providing objective, factual story summaries. It is programmed to never generate subjective, offensive, or harmful commentary about characters or themes.
- Ensuring dignity and user autonomy: The user is always in control. The companion can be easily disabled, muted, or customized. It acts as a helpful tool, never forcing interactions.

e) Privacy & Security

- Ensuring data privacy: Video frames and audio used to understand the screen are processed and immediately discarded. We do not store personal viewing habits or sell user data.
- Avoiding harmful use: The system is built strictly as a viewer aid and includes safeguards to ensure it cannot be manipulated to copy, record, or distribute copyrighted streaming content.

Governance & Compliance

- Internal governance rules (who approves launch/changes): All major updates, new feature launches, or changes to how user data is handled require unanimous approval from the core team leadership. Any changes affecting the "Spoiler Shield" require rigorous peer testing before deployment.
- Standards & frameworks you reference: We align with the OECD Principles on AI and UNESCO's Ethics of AI, specifically focusing on transparency, fairness, and ensuring the AI respects human rights and privacy.
- Audit & monitoring plans (ongoing compliance): We will conduct monthly reviews of user error reports (flagged spoilers or incorrect lore). We will also run automated weekly tests on the database to ensure new show updates do not break the privacy or spoiler-protection rules.

Scope & Limitations of the MVP

Purpose: Explain what your MVP will and will not do.

When: by Week 5

Tip: At the end, you only have 10 weeks. You will need to prioritise –focus on the 2-3 functionalities that will deliver the greatest user value for your MVP.

MVP Scope

- Video-Synced Chat Interface: A web application featuring a video player and a chat box. When the user pauses the video to ask a question, the app extracts the exact timestamp.
- Single-Episode "Vault": A pre-built database containing timestamped scene summaries and subtitle data for one specific test episode.
- The Spoiler Shield: A functional governance filter that physically blocks the AI from accessing any story data timestamped after the user's exact pause point.
- User Value Focus: Proving the AI can successfully filter future events (the Spoiler Shield) and provide context based on past dialogue is our core innovation. Focusing on this text-based temporal logic delivers the greatest proof of concept within the timeframe.

Limitations

What are the known constraints of your MVP?

- No Visual Processing (Text-Only Context): The AI relies entirely on subtitle data and text summaries mapped to timestamps. It does not possess computer vision to analyze the actual video frames. As a result, the user cannot ask visual-dependent questions (e.g., "Who is the man standing in the background?" or "What is the symbol on that shield?"). The system can only clarify events, characters, or lore explicitly related to the text data.
- No Streaming Integration: The solution is a standalone web application and is not directly embedded into commercial platforms like Netflix or Smart TV operating systems.
- Restricted Knowledge Base: The AI is pre-loaded with the structured lore of only one specific test episode, or might not be full structured lore but AI generated summary of each subtitle that give it some content, the answer not necessarily very good since we skipped the agent that build a the knowledge graph and it cannot answer questions about the broader series or other shows.

What is next

- Phase 1 (Stress Testing): Conduct user testing where participants actively try to "trick" the chatbot into revealing a spoiler to validate the Shield's logic.
- Phase 2 (Visual Perception): Integrate the "Perception Agent" (computer vision) so the tool can identify unmentioned characters and objects directly from the video frames, removing the text-only limitation.
- Phase 3 (Ecosystem Scale): Expand the database to encompass full-series wikis and deploy the front-end as a browser extension that overlays directly onto streaming web players.



Checkpoint –Week 5

Tip: Validate technical decisions early with prototypes and stress tests. Ensure ethical and governance considerations are not an afterthought—they should be integrated into design from the start.

At this stage, you should be clear about the MVP that you are building and will submit at the end of the this program.

- ☐ Is the architecture modular, scalable, and aligned with solution goals? Does it supports performance, scalability, and maintainability?
- ☐ Are workflows clear, efficient, and user-centric? Does it aligns with user experience and business objectives?
- ☐ Are tools and integrations secure, compitble (interoperable), and cost-effective?
- ☐ Is the intelligence logic well-thought and justified?
- ☐ Does the design ensures data privacy, fairness, explainability, and bias mitigation?
- ☐ Does the governance framework addresses compliance, and accountability?

Open Questions & Assumptions (for now this is based on the MVP)

What we don't know yet

- Context Limits: We do not know if the AI can accurately explain complex character relationships using only brief text summaries, without the visual context of the scene.
- Search Accuracy: It is unclear if basic keyword searching will reliably retrieve the correct past summary chunk when a user asks a vague question (e.g., "Why did he do that?").
- UX Friction: We don't know if users will find manually adjusting the "Current Time" slider too tedious compared to an automated video player.

Assumptions we are making

- Context Limits: We do not know if the AI can accurately explain complex character relationships using only brief text summaries, without the visual context of the scene.
- Search Accuracy: It is unclear if basic keyword searching will reliably retrieve the correct past summary chunk when a user asks a vague question (e.g., "Why did he do that?").
- UX Friction: We don't know if users will find manually adjusting the "Current Time" slider too tedious compared to an automated video player.

Validation research

- Spoiler Stress Test (Red Teaming): We will actively try to "trick" the AI with questions about the episode's ending while the slider is set to Minute 5, verifying the shield holds.
- Retrieval Quality Test: We will ask indirect or ambiguous questions to verify the backend logic retrieves the correct historical text chunk before passing it to the AI.
- UX Flow Test: We will have a test user watch a 10-minute clip on a separate screen and attempt to use our chat interface alongside it to see if the manual slider feels natural.

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Continue the journey

- Weeks 6-8:
 - Implement the technical work
 - Ensure depth & robustness
 - Carry out user testing & iteration
- Weeks 9-10: Prepare and finalise your project submission

Good Luck!!